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Assignment Cover Page- Jan'26 Session

Maximum Marks: 30,
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Course Name: Classcial Mechanics ………………………..

Course Code: MMM-6112 ………………………..

Date of Submission: ……………………….

Note- The assignment Question have 5 Sections/Blocks. Kindly attempt Any ONE question from Each of the 5 blocks. Each question carries equal marks. For Eg:

Block 1 Attempt- Question 1a or 1b

Block 2 Attempt- Question 2a or 2b

Block 3 Attempt- Question 3a or 3b

Block 4 Attempt- Question 4a or 4b

Block 5 Attempt- Question 5a or 5b

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Type your assignment here…

or

You may attach handwritten assignment also (in PDF format)

Q1 B: Given a square matrix A, prove analytically that the trace of A is equal to the sum of its eigenvalues. Provide a numerical example to illustrate this property using a specific matrix.

Radhika Sharma

1 Definitions

Definition 1 (Trace)**.** The **trace** of an $n \times n$ square matrix $A = (a_{ij})$ is the sum of its main diagonal entries:

$$\mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Definition 2 (Eigenvalue)**.** A scalar $\lambda \in \mathbb{C}$ is an **eigenvalue** of A if there exists a non-zero vector $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$. The eigenvalues are exactly the roots of the **characteristic polynomial**

$$p(\lambda) = \det(\lambda I - A).$$

2 The Characteristic Polynomial and Vieta's Formulas

Lemma 3 (Structure of the Characteristic Polynomial)**.** *For an $n \times n$ matrix A with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (counted with algebraic multiplicity), the characteristic polynomial is*

$$p(\lambda) = \det(\lambda I - A) = (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n) = \lambda^n - \left(\sum_{i=1}^{n} \lambda_i\right)\lambda^{n-1} + \cdots + (-1)^n \det(A).$$

In particular, the coefficient of λ^{n-1} is $-\sum_{i=1}^{n} \lambda_i$.

Proof. Since the eigenvalues are precisely the roots of $p(\lambda)$ and p is monic of degree n, the factored form follows from the Fundamental Theorem of Algebra. Expanding the product $(\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n)$, the coefficient of λ^{n-1} is $-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$, which is Vieta's formula for the sum of the roots. $\square$

3 Main Theorem: $\operatorname{tr}(A) = \sum \lambda_i$

Theorem 4. *Let A be an $n \times n$ matrix over $\mathbb{C}$ with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (counted with algebraic multiplicity). Then*

$$\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

Proof. We expand $p(\lambda) = \det(\lambda I - A)$ directly and compare the coefficient of λ^{n-1} with that obtained from the factored form in Lemma 3.

Step 1: Expand $\det(\lambda I - A)$.
Write

$$\lambda I - A = \begin{pmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{pmatrix}.$$

The determinant is a sum over all permutations $\sigma \in S_n$:

$$\det(\lambda I - A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} (\lambda I - A)_{i,\sigma(i)}.$$

Step 2: Identify which terms contribute to λ^{n-1}.
Each factor $(\lambda I - A)_{i,\sigma(i)}$ equals:

- $(\lambda - a_{ii})$ if $\sigma(i) = i$ (diagonal entry), contributing a λ or a constant;
- $-a_{i,\sigma(i)}$ if $\sigma(i) \neq i$ (off-diagonal entry), a constant.

A term $\prod_{i=1}^{n} (\lambda I - A)_{i,\sigma(i)}$ has degree k in λ, where k is the number of fixed points of σ (positions where $\sigma(i) = i$).

To obtain λ^{n-1}, we need *exactly* $n-1$ *fixed points.* But a permutation with $n-1$ fixed points must fix all n points (since fixing $n-1$ forces the last to be fixed too). Therefore the *only* permutation contributing to λ^{n-1} is the **identity** $\sigma = \mathrm{id}$.

Step 3: Compute the contribution of the identity permutation.
For $\sigma = \mathrm{id}$, $\operatorname{sgn}(\mathrm{id}) = 1$ and

$$\prod_{i=1}^{n} (\lambda - a_{ii}) = \lambda^n - (a_{11} + a_{22} + \cdots + a_{nn})\lambda^{n-1} + \cdots = \lambda^n - \operatorname{tr}(A)\,\lambda^{n-1} + \cdots$$

All other permutations $\sigma \neq \mathrm{id}$ have at most $n-2$ fixed points, so they contribute terms of degree at most $n-2$ in λ.

Step 4: Equate coefficients.
From Steps 2–3, the coefficient of λ^{n-1} in $p(\lambda) = \det(\lambda I - A)$ is $-\operatorname{tr}(A)$.
From Lemma 3, the coefficient of λ^{n-1} in $(\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$ is $-\sum_{i=1}^{n} \lambda_i$.
Therefore:

$$-\operatorname{tr}(A) = -\sum_{i=1}^{n} \lambda_i \implies \operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i. \qquad \square$$

Remark. The same argument shows that $(-1)^n \det(A)$ is the constant term of $p(\lambda)$, i.e. $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$.

4 Explicit Verification for 2×2 and 3×3 Matrices

4.1 The 2×2 Case

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$\det(\lambda I - A) = (\lambda - a)(\lambda - d) - bc = \lambda^2 - (a + d)\lambda + (ad - bc).$$

By Vieta's formulas: $\lambda_1 + \lambda_2 = a + d = \operatorname{tr}(A)$. ✓

4.2 The 3×3 Case

Let $A = (a_{ij})_{3\times 3}$. The diagonal product gives

$$(\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33}) = \lambda^3 - (a_{11} + a_{22} + a_{33})\lambda^2 + \cdots$$

Every other term in $\det(\lambda I - A)$ involves at most one diagonal factor, contributing at most λ^1. Hence the λ^2 coefficient is exactly $-(a_{11} + a_{22} + a_{33}) = -\operatorname{tr}(A)$, confirming $\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr}(A)$. ✓

5 Numerical Examples

5.1 Example 1: 2×2 Matrix

Example 5. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Trace:

$$\operatorname{tr}(A) = 1 + 4 = 5.$$

Characteristic polynomial:

$$p(\lambda) = \det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{vmatrix} = (\lambda - 1)(\lambda - 4) - 6 = \lambda^2 - 5\lambda - 2.$$

Eigenvalues (quadratic formula):

$$\lambda = \frac{5 \pm \sqrt{25 + 8}}{2} = \frac{5 \pm \sqrt{33}}{2}.$$

Sum of eigenvalues (Vieta's formula directly from the polynomial):

$$\lambda_1 + \lambda_2 = 5 = \operatorname{tr}(A). \quad ✓$$

5.2 Example 2: 3×3 Matrix

Example 6. Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

(Upper triangular matrix — eigenvalues are the diagonal entries.)
Trace:

$$\operatorname{tr}(A) = 2 + 3 + 5 = 10.$$

Characteristic polynomial: Since A is upper triangular,

$$p(\lambda) = (\lambda - 2)(\lambda - 3)(\lambda - 5).$$

Eigenvalues:

$$\lambda_1 = 2, \quad \lambda_2 = 3, \quad \lambda_3 = 5.$$

Sum of eigenvalues:

$$\lambda_1 + \lambda_2 + \lambda_3 = 2 + 3 + 5 = 10 = \operatorname{tr}(A). \quad \checkmark$$

5.3 Example 3: 3×3 Matrix with Non-Trivial Eigenvalues

Example 7. Let

$$A = \begin{pmatrix} 4 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Trace:

$$\operatorname{tr}(A) = 4 + 3 + 2 = 9.$$

Characteristic polynomial: Upper triangular, so

$$p(\lambda) = (\lambda - 4)(\lambda - 3)(\lambda - 2).$$

Eigenvalues:

$$\lambda_1 = 4, \quad \lambda_2 = 3, \quad \lambda_3 = 2.$$

Sum of eigenvalues:

$$4 + 3 + 2 = 9 = \operatorname{tr}(A). \quad \checkmark$$

6 Summary

Matrix	**Size**	$\operatorname{tr}(A)$	**Eigenvalues**	$\sum \lambda_i$
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	2×2	5	$\dfrac{5 \pm \sqrt{33}}{2}$	5
$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 5 \end{pmatrix}$	3×3	10	$2,\ 3,\ 5$	10
$\begin{pmatrix} 4 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \end{pmatrix}$	3×3	9	$4,\ 3,\ 2$	9

Conclusion. The proof proceeds in two stages:

1. Expand $\det(\lambda I - A)$ and show the coefficient of λ^{n-1} is $-\operatorname{tr}(A)$ (only the identity permutation contributes).
2. Factor $\det(\lambda I - A) = \prod(\lambda - \lambda_i)$ and apply Vieta's formula: the coefficient of λ^{n-1} is $-\sum \lambda_i$.

Equating the two expressions gives $\operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i$.

Q2A: Derive analytically the property of Fourier transform for the derivative of a function. Apply this property to find the Fourier transform of a specific function and its derivative, providing numerical calculations to support the result.

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Part 1: Derivation of the Derivative Property

Definition

The Fourier transform of a function $f(x)$ is defined as

$$\hat{f}(k) = \mathcal{F}[f](k) = \int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx.$$

Theorem 1 (Derivative Property)**.** *Let f be a continuously differentiable, absolutely integrable function on $\mathbb{R}$ with $\lim_{x\to\pm\infty} f(x) = 0$. Then*

$$\mathcal{F}[f'](k) = -ik\, \hat{f}(k).$$

Proof. By definition,

$$\mathcal{F}[f']\,(k) = \int_{-\infty}^{\infty} f'(x)\, e^{ikx}\, dx.$$

Integrate by parts with $u = e^{ikx}$ and $dv = f'(x)\, dx$, so $du = ik\, e^{ikx}\, dx$ and $v = f(x)$:

$$\int_{-\infty}^{\infty} f'(x)\, e^{ikx}\, dx = \Big[f(x)\, e^{ikx}\Big]_{-\infty}^{\infty} - ik \int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx.$$

Boundary term vanishes. Since $|e^{ikx}| = 1$ for real k and $\lim_{x\to\pm\infty} f(x) = 0$,

$$\Big[f(x)\, e^{ikx}\Big]_{-\infty}^{\infty} = 0.$$

Identify the remaining integral as $\hat{f}(k)$:

$$\mathcal{F}[f']\,(k) = -ik \int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx = -ik\, \hat{f}(k). \qquad \square$$

Remark. Applying the property n times gives the general result $\mathcal{F}[f^{(n)}](k) = (-ik)^n\, \hat{f}(k)$. In particular, $\mathcal{F}[f''](k) = -k^2\, \hat{f}(k)$.

Part 2: Application to a Specific Function

Choice of function

Let

$$f(x) = \begin{cases} e^{-ax}, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad a > 0.$$

This function is absolutely integrable, decays to zero as $x \to +\infty$, and vanishes identically for $x < 0$.

Step 1: Compute $\hat{f}(k)$ directly

$$\hat{f}(k) = \int_0^\infty e^{-ax}\, e^{ikx}\, dx = \int_0^\infty e^{-(a-ik)x}\, dx.$$

Since $a > 0$ ensures convergence:

$$\hat{f}(k) = \left[\frac{e^{-(a-ik)x}}{-(a-ik)} \right]_0^\infty = 0 - \frac{1}{-(a-ik)} = \frac{1}{a-ik}.$$

Step 2: Find $\mathcal{F}[f'](k)$ using the derivative property

By the theorem:

$$\mathcal{F}[f'](k) = -ik\, \hat{f}(k) = \frac{-ik}{a-ik}.$$

Step 3: Verify by computing $\mathcal{F}[f'](k)$ directly

For $x > 0$, $f'(x) = -ae^{-ax}$. At $x = 0$, the function jumps from 0 to 1 (since $f(0^-) = 0$ and $f(0^+) = 1$), contributing a Dirac delta term. In the distributional sense:

$$f'(x) = -ae^{-ax}\, \mathbf{1}_{x \geq 0} + \delta(x).$$

Taking the Fourier transform:

$$\begin{aligned} \mathcal{F}[f']\,(k) &= \int_0^\infty (-a)\, e^{-ax}\, e^{ikx}\, dx + \int_{-\infty}^\infty \delta(x)\, e^{ikx}\, dx \\ &= -a \cdot \frac{1}{a-ik} + 1 \\ &= \frac{-a + (a-ik)}{a-ik} = \frac{-ik}{a-ik}. \end{aligned}$$

This matches the result from Step 2 exactly. ✓

Part 3: Numerical Verification

Set $a = 2$ and $k = 1$.

Step A: Compute $\hat{f}(1)$

$$\hat{f}(1) = \frac{1}{2-i} = \frac{2+i}{2^2+1^2} = \frac{2+i}{5} = 0.4 + 0.2i.$$

Step B: Apply derivative property

$$-ik\,\hat{f}(k)\big|_{k=1} = -i(1)(0.4+0.2i) = -0.4i - 0.2i^2 = 0.2 - 0.4i.$$

Step C: Compute $\mathcal{F}[f'](1)$ directly

$$\left.\frac{-ik}{a-ik}\right|_{a=2,\,k=1} = \frac{-i}{2-i} = \frac{-i(2+i)}{5} = \frac{-2i+1}{5} = 0.2 - 0.4i.$$

Result: Both methods give $\mathcal{F}[f'](1) = 0.2 - 0.4i$. ✓

Verify also at $k = 2$, $a = 2$:

$$\hat{f}(2) = \frac{1}{2-2i} = \frac{2+2i}{8} = 0.25 + 0.25i.$$

$$-ik\,\hat{f}(k)\big|_{k=2} = -2i(0.25+0.25i) = -0.5i - 0.5i^2 = 0.5 - 0.5i.$$
$$\left.\frac{-ik}{a-ik}\right|_{a=2,\,k=2} = \frac{-2i}{2-2i} = \frac{-2i(2+2i)}{8} = \frac{4-4i}{8} = 0.5 - 0.5i. \quad ✓$$

Conclusion

The derivative property $\mathcal{F}[f'](k) = -ik\,\hat{f}(k)$ was:

1. **Derived analytically** via integration by parts, with the boundary term vanishing due to the decay condition on f.
2. **Applied** to $f(x) = e^{-ax}\mathbf{1}_{x\geq 0}$, giving $\mathcal{F}[f'](k) = -ik/(a-ik)$.
3. **Verified numerically** at $(a,k) = (2,1)$ and $(2,2)$, with both the direct and property-based methods agreeing exactly.

Q3A: Derive Hamilton's equations of motion for a particle in a conservative force field using Hamilton's principle.

Radhika Sharma

1. Setup and Definitions

Consider a particle of mass m moving in three dimensions under a conservative force $\mathbf{F} = -\nabla V(\mathbf{q})$, where V is the potential energy and $\mathbf{q} = (q_1, q_2, q_3)$ are generalised coordinates.

Definition (Lagrangian)**.** The **Lagrangian** is

$$L(\mathbf{q}, \dot{\mathbf{q}}) = T - V = \frac{1}{2} m \sum_{i=1}^{3} \dot{q}_i^2 - V(\mathbf{q}),$$

where T is the kinetic energy and dots denote time derivatives.

Definition (Generalised Momentum)**.** The **generalised (conjugate) momentum** corresponding to q_i is

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = m\dot{q}_i.$$

The pair (q_i, p_i) forms a set of *canonical variables.*

Definition (Hamiltonian)**.** The **Hamiltonian** is the Legendre transform of L with respect to $\dot{\mathbf{q}}$:

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^{3} p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}),$$

where $\dot{q}_i$ is expressed in terms of p_i via $\dot{q}_i = p_i/m$. For our particle,

$$H = \sum_i p_i \cdot \frac{p_i}{m} - \left(\frac{1}{2m} \sum_i p_i^2 - V \right) = \frac{|\mathbf{p}|^2}{2m} + V(\mathbf{q}) = T + V.$$

The Hamiltonian equals the total mechanical energy.

2. Hamilton's Principle (Principle of Stationary Action)

Definition (Action Functional)**.** The **action** along a path $\mathbf{q}(t)$ between fixed endpoints $\mathbf{q}(t_1)$ and $\mathbf{q}(t_2)$ is

$$S[\mathbf{q}] = \int_{t_1}^{t_2} L(\mathbf{q}, \dot{\mathbf{q}})\, dt.$$

Hamilton's Principle states: the physical trajectory of the system is the path $\mathbf{q}(t)$ for which the action S is *stationary*, i.e. $\delta S = 0$, subject to fixed endpoints $\delta\mathbf{q}(t_1) = \delta\mathbf{q}(t_2) = 0$.

3. From Lagrangian to Hamiltonian Formulation

We rewrite the action in terms of the canonical variables $(\mathbf{q}, \mathbf{p})$ by substituting $L = \sum_i p_i \dot{q}_i - H$:

$$S = \int_{t_1}^{t_2} \Big(\sum_{i=1}^{3} p_i \dot{q}_i - H(\mathbf{q}, \mathbf{p}) \Big)\, dt.$$

We now treat $\mathbf{q}(t)$ and $\mathbf{p}(t)$ as *independent* variations.

4. Derivation of Hamilton's Equations

Theorem 1 (Hamilton's Equations of Motion). *The stationarity condition $\delta S = 0$ under independent variations δq_i and δp_i (with $\delta q_i(t_1) = \delta q_i(t_2) = 0$) yields*

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, 2, 3.$$

Proof. Compute the first variation of S:

$$\delta S = \int_{t_1}^{t_2} \sum_i \Big(\delta p_i\, \dot{q}_i + p_i\, \delta\dot{q}_i - \frac{\partial H}{\partial q_i}\delta q_i - \frac{\partial H}{\partial p_i}\delta p_i \Big)\, dt.$$

Step 1 — Integrate the $p_i\, \delta\dot{q}_i$ term by parts.

$$\int_{t_1}^{t_2} p_i\, \delta\dot{q}_i\, dt = \Big[p_i\, \delta q_i \Big]_{t_1}^{t_2} - \int_{t_1}^{t_2} \dot{p}_i\, \delta q_i\, dt.$$

The boundary term vanishes since $\delta q_i(t_1) = \delta q_i(t_2) = 0$.

Step 2 — Collect terms.

Substituting back:

$$\delta S = \int_{t_1}^{t_2} \sum_i \left[\left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right) \delta p_i - \left(\dot{p}_i + \frac{\partial H}{\partial q_i} \right) \delta q_i \right] dt = 0.$$

Step 3 — Apply the Fundamental Lemma of Calculus of Variations.

Since δq_i and δp_i are *independent and arbitrary*, each coefficient must vanish identically:

$$\delta p_i \text{ arbitrary:} \quad \dot{q}_i - \frac{\partial H}{\partial p_i} = 0 \implies \boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}},$$

$$\delta q_i \text{ arbitrary:} \quad \dot{p}_i + \frac{\partial H}{\partial q_i} = 0 \implies \boxed{\dot{p}_i = -\frac{\partial H}{\partial q_i}}.$$

These are Hamilton's canonical equations of motion. □

5. Explicit Form for a Particle in a Conservative Field

For $H = \dfrac{|\mathbf{p}|^2}{2m} + V(\mathbf{q})$, the partial derivatives are

$$\frac{\partial H}{\partial p_i} = \frac{p_i}{m}, \qquad \frac{\partial H}{\partial q_i} = \frac{\partial V}{\partial q_i}.$$

Hamilton's equations therefore give:

$$\dot{q}_i = \frac{p_i}{m} \quad \Longleftrightarrow \quad p_i = m\dot{q}_i \quad \text{(momentum definition)},$$

$$\dot{p}_i = -\frac{\partial V}{\partial q_i} \quad \Longleftrightarrow \quad m\ddot{q}_i = -\frac{\partial V}{\partial q_i} = F_i \quad \text{(Newton's second law)}.$$

The two first-order equations together reproduce Newton's second law $m\ddot{\mathbf{q}} = -\nabla V$, confirming the derivation is consistent.

6. Conservation of Energy

Using Hamilton's equations, the total time derivative of H is

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial p_i}\dot{p}_i \right) + \frac{\partial H}{\partial t} = \sum_i \left(\frac{\partial H}{\partial q_i} \cdot \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \cdot \frac{\partial H}{\partial q_i} \right) + \frac{\partial H}{\partial t}.$$

The sum cancels identically, leaving

$$\frac{dH}{dt} = \frac{\partial H}{\partial t}.$$

For a conservative field $V = V(\mathbf{q})$ (no explicit time dependence), $\partial H/\partial t = 0$, so

$$\frac{dH}{dt} = 0 \implies H = E = \text{constant}.$$

The Hamiltonian (total energy) is conserved along every physical trajectory.

7. Summary

Starting from Hamilton's principle $\delta S = 0$ with action $S = \int_{t_1}^{t_2} (\sum_i p_i\dot{q}_i - H)\, dt$, independent variation of $\mathbf{q}$ and $\mathbf{p}$ and integration by parts yields the **canonical equations**:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

For a particle in a conservative field with $H = |\mathbf{p}|^2/2m + V(\mathbf{q})$, these reduce to Newton's second law. The formulation replaces one second-order ODE per coordinate with two first-order ODEs in the canonical pair (q_i, p_i), and reveals the deep symmetry between coordinates and momenta that underlies classical mechanics.

Q4B: Calculate the relativistic kinetic energy of a particle with a given velocity.

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Relativistic Kinetic Energy of a Particle

Setup

Let a particle of rest mass m_0 move with velocity v relative to an inertial frame. We wish to derive and calculate its relativistic kinetic energy.

Key Definitions

The **Lorentz factor** is defined as:

$$\gamma = \frac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

where $c = 3 \times 10^8$ m/s is the speed of light in vacuum.

Derivation

The **relativistic total energy** of the particle is:

$$E = \gamma m_0 c^2$$

The **rest energy** (energy when $v = 0$, i.e., $\gamma = 1$) is:

$$E_0 = m_0 c^2$$

The **relativistic kinetic energy** K is the difference between the total energy and the rest energy:

$$\boxed{K = E - E_0 = (\gamma - 1)\, m_0 c^2}$$

Explicit Form

Substituting γ:

$$K = \left(\frac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}} - 1 \right) m_0 c^2$$

Classical Limit Verification

In the non-relativistic limit $v \ll c$, we expand γ using the binomial approximation:

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + O\left(\frac{v^4}{c^4}\right)$$

Therefore:

$$K \approx \left(1 + \frac{1}{2}\frac{v^2}{c^2} - 1\right) m_0 c^2 = \frac{1}{2} m_0 v^2$$

which recovers the classical Newtonian kinetic energy, as expected.

Numerical Example

Let $m_0 = 9.109 \times 10^{-31}$ kg (electron rest mass) and $v = 0.8c$.

Compute γ:

$$\gamma = \frac{1}{\sqrt{1-(0.8)^2}} = \frac{1}{\sqrt{1-0.64}} = \frac{1}{\sqrt{0.36}} = \frac{1}{0.6} = \frac{5}{3} \approx 1.6667$$

Compute the rest energy:

$$E_0 = m_0 c^2 = (9.109 \times 10^{-31})(3 \times 10^8)^2 = 8.198 \times 10^{-14} \text{ J}$$

Compute the kinetic energy:

$$K = (\gamma - 1)\, m_0 c^2 = \left(\frac{5}{3} - 1\right)(8.198 \times 10^{-14}) = \frac{2}{3} \times 8.198 \times 10^{-14} \approx 5.465 \times 10^{-14} \text{ J}$$

Result

$$\boxed{K = (\gamma - 1)\, m_0 c^2 \approx 5.465 \times 10^{-14} \text{ J}}$$

for an electron travelling at $v = 0.8c$.

Q5B: Calculate the relativistic momentum and energy of a particle undergoing acceleration.

Radhika Sharma

Q5B: Relativistic Momentum and Energy of an Accelerating Particle

Setup

Consider a particle of rest mass m_0 undergoing **constant proper acceleration** α, starting from rest at $t = 0$ in the lab frame S. Proper acceleration is the acceleration measured in the instantaneous rest frame of the particle, and is *not* equivalent to a constant lab-frame force.

Proper Acceleration: Correct Definition

The 4-acceleration is defined as:

$$A^\mu = \frac{dU^\mu}{d\tau}$$

where $U^\mu = \gamma(c, \mathbf{v})$ is the 4-velocity and τ is proper time.

The magnitude of the 4-acceleration is the proper acceleration α:

$$A^\mu A_\mu = -\alpha^2 \qquad \text{(using signature } +---\text{)}$$

In 1+1 dimensions this gives:

$$\gamma^4 \left(\frac{dv}{dt}\right)^2 \cdot \frac{1}{c^2} \cdot c^2 = \alpha^2 \implies \gamma^3 \frac{dv}{dt} = \alpha$$

Hence the lab-frame force is *not* constant:

$$F = \frac{dp}{dt} = \frac{d(\gamma m_0 v)}{dt} = m_0 \gamma^3 \frac{dv}{dt} \cdot \frac{1}{\gamma^2}$$

More precisely, from $p = \gamma m_0 v$ and the chain rule:

$$\frac{dp}{dt} = m_0 \gamma^3 \frac{dv}{dt} = m_0 \alpha \gamma^2$$

This is velocity-dependent and **not constant**, unlike the previous (incorrect) treatment.

Velocity as a Function of Coordinate Time

We integrate the proper acceleration equation. Since $\gamma^3 \dfrac{dv}{dt} = \alpha$, write:

$$\frac{dv}{dt} = \frac{\alpha}{\gamma^3} = \alpha \left(1 - \frac{v^2}{c^2}\right)^{3/2}$$

Separating variables:

$$\int_0^v \frac{dv'}{(1 - v'^2/c^2)^{3/2}} = \alpha t$$

The left-hand side evaluates to:

$$\frac{v}{\sqrt{1 - v^2/c^2}} = \alpha t$$

Solving for $v(t)$:

$$\boxed{v(t) = \frac{\alpha t}{\sqrt{1 + \dfrac{\alpha^2 t^2}{c^2}}}}$$

As $t \to \infty$, $v(t) \to c$, confirming the speed of light is never reached.

Lorentz Factor

$$\gamma(t) = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{1}{\sqrt{1 - \dfrac{\alpha^2 t^2}{c^2 + \alpha^2 t^2}}} = \sqrt{\frac{c^2 + \alpha^2 t^2}{c^2}}$$

$$\boxed{\gamma(t) = \sqrt{1 + \frac{\alpha^2 t^2}{c^2}}}$$

Relativistic Momentum

$$p(t) = \gamma(t)\, m_0\, v(t) = \sqrt{1 + \frac{\alpha^2 t^2}{c^2}} \cdot m_0 \cdot \frac{\alpha t}{\sqrt{1 + \alpha^2 t^2/c^2}}$$

$$\boxed{p(t) = m_0 \alpha t}$$

Note: although $p(t) = m_0 \alpha t$ resembles the Newtonian result, it arises here from integrating the proper-acceleration equation of motion, *not* from a constant force assumption. The lab-frame force $F = dp/dt = m_0 \alpha$ happens to be constant as a consequence of the algebra, but the physical origin is proper acceleration, not a prescribed constant force.

Relativistic Total Energy

$$\boxed{E(t) = \gamma(t)\, m_0 c^2 = m_0 c^2 \sqrt{1 + \frac{\alpha^2 t^2}{c^2}}}$$

Relativistic Kinetic Energy

$$K(t) = E(t) - m_0 c^2 = m_0 c^2 \left(\sqrt{1 + \frac{\alpha^2 t^2}{c^2}} - 1 \right)$$

Proper Time

Integrating $d\tau = dt/\gamma(t)$:

$$\tau(t) = \int_0^t \frac{dt'}{\gamma(t')} = \int_0^t \frac{dt'}{\sqrt{1 + \alpha^2 t'^2/c^2}} = \frac{c}{\alpha} \sinh^{-1}\left(\frac{\alpha t}{c}\right)$$

Inverting: $t = \dfrac{c}{\alpha} \sinh\left(\dfrac{\alpha \tau}{c}\right)$, so the worldline in terms of proper time is:

$$x(\tau) = \frac{c^2}{\alpha}\left(\cosh \frac{\alpha \tau}{c} - 1\right), \qquad t(\tau) = \frac{c}{\alpha} \sinh \frac{\alpha \tau}{c}$$

which is the standard **hyperbolic motion** of uniformly accelerated relativistic particles.

Energy–Momentum Relation Verification

$$E^2 - (pc)^2 = m_0^2 c^4 \left(1 + \frac{\alpha^2 t^2}{c^2}\right) - m_0^2 \alpha^2 t^2 c^2 = m_0^2 c^4 \quad \checkmark$$

Classical Limit

For $\alpha t \ll c$:

$$\gamma(t) \approx 1 + \frac{\alpha^2 t^2}{2c^2}, \quad v(t) \approx \alpha t, \quad K(t) \approx \tfrac{1}{2} m_0 \alpha^2 t^2 = \tfrac{1}{2} m_0 v^2 \quad \checkmark$$

Numerical Example

Let $m_0 = 1.673 \times 10^{-27}$ kg (proton), $\alpha = 10^{16}$ m/s^2, $t = 10^{-8}$ s.

$$\alpha t = 10^8 \text{ m/s}, \qquad \frac{\alpha t}{c} = \frac{10^8}{3 \times 10^8} = \tfrac{1}{3}$$

$$\gamma = \sqrt{1 + \tfrac{1}{9}} = \sqrt{\tfrac{10}{9}} \approx 1.054$$

$$p = m_0 \alpha t = (1.673 \times 10^{-27})(10^8) = 1.673 \times 10^{-19} \text{ kg m/s}$$

$$E = \gamma m_0 c^2 \approx 1.054 \times (1.673 \times 10^{-27})(9 \times 10^{16}) \approx 1.586 \times 10^{-10} \text{ J}$$

$$K = (\gamma - 1) m_0 c^2 \approx 0.054 \times 1.506 \times 10^{-10} \approx 8.13 \times 10^{-12} \text{ J}$$

$$\tau = \frac{c}{\alpha} \sinh^{-1}\!\left(\tfrac{1}{3}\right) = \frac{3 \times 10^8}{10^{16}} \sinh^{-1}(0.333) \approx 3 \times 10^{-8} \times 0.3272 \approx 9.82 \times 10^{-9} \text{ s}$$

Summary Table

Quantity	**Expression (constant proper acceleration α)**
Velocity	$v(t) = \dfrac{\alpha t}{\sqrt{1 + \alpha^2 t^2 / c^2}}$
Lorentz factor	$\gamma(t) = \sqrt{1 + \dfrac{\alpha^2 t^2}{c^2}}$
Relativistic momentum	$p(t) = m_0 \alpha t$
Total energy	$E(t) = m_0 c^2 \sqrt{1 + \dfrac{\alpha^2 t^2}{c^2}}$
Kinetic energy	$K(t) = m_0 c^2 \left(\sqrt{1 + \dfrac{\alpha^2 t^2}{c^2}} - 1\right)$
Proper time	$\tau = \dfrac{c}{\alpha} \sinh^{-1}\!\left(\dfrac{\alpha t}{c}\right)$